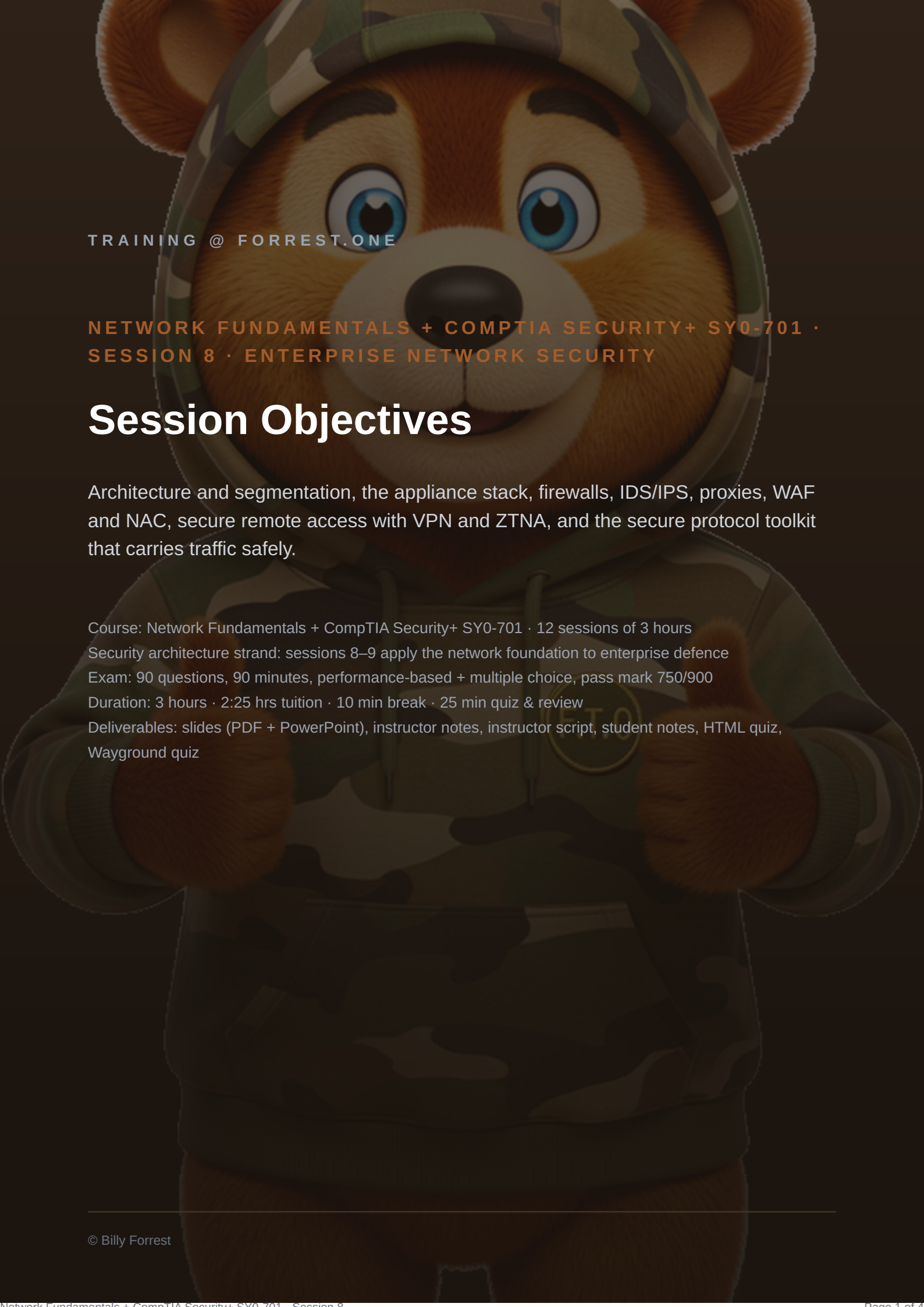




TRAINING @ FORREST.ONE

**NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 8 · ENTERPRISE NETWORK SECURITY**

Session Objectives

Architecture and segmentation, the appliance stack, firewalls, IDS/IPS, proxies, WAF and NAC, secure remote access with VPN and ZTNA, and the secure protocol toolkit that carries traffic safely.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Security architecture strand: sessions 8–9 apply the network foundation to enterprise defence

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Session Objectives — Enterprise Network Security

This session applies the network foundation from sessions 1–3 to enterprise defence, and follows the expanded narration spine: the physical and human layer, segmenting the network, guests and personal devices, hardening and admission control, keys and content control, and trust zones. The exam-objective strand (8A architecture, 8B appliances, 8C remote access, 8D secure protocols) is retained and consolidated into exam-reference slides, because scenario questions reward exactly that: pick the right control and place it correctly. Defence in depth runs throughout: no single layer carries the whole burden.

Learning outcomes

By the end of the 3-hour session, candidates will be able to:

- Explain physical controls for equipment rooms, wiring closets and data centres, and compare key, combination, electronic and biometric access.
- Describe surveillance and visitor controls, including camera coverage, retention policy and access logging.
- Compare low and high interaction honeypots, explain what a honeynet exposes, and state why isolation and lawful consent are mandatory.
- Explain what an audit tests, the difference between internal audit and external third party assessment, and how to prepare and remediate.
- State the demands of PCI DSS, GDPR and data locality, and explain why logs and configurations are the evidence of compliance.
- Explain segmentation principles: VLANs per department, firewall zones, micro-segmentation, and planning from data sensitivity and trust level.
- Contain internet of things and industrial control systems, using the Purdue model and air gaps where intrusion risk must be near zero.
- Justify an availability-first approach in operational technology, where devices cannot be patched or rebooted freely.
- Design a guest network with its own segment, no internal route, a bandwidth cap and a captive portal.
- Define a bring your own device programme: compliance definition, enrolment, certificates, data separation and selective wipe.
- Harden devices by removing unused services, ports and protocols, and by replacing default credentials.
- Build a secure configuration baseline from published guides, then verify compliance and catch configuration drift.
- Explain network access control posture assessment, its allow, quarantine and deny outcomes, and the role of a remediation network.

- Describe 802.1X port based authentication, the supplicant, authenticator and RADIUS server roles, and phased deployment.
- Describe the key lifecycle (generation, distribution, storage, rotation, revocation, destruction) and where keys should live.
- Explain access control list ordering, URL and DNS filtering, content filtering, deep packet inspection and data loss prevention.
- Describe trust zones, the screened subnet, zone based policy with deliberate asymmetry, defence in depth and zero trust.
- Compare IDS and IPS, firewall types, and remote access options including VPN flavours and ZTNA.
- State the jobs of TLS/HTTPS, SSH, IPsec, S/MIME, DNSSEC and HSTS, and the certificate validation steps that defeat man-in-the-middle attacks.

Course context

Item	Detail
Course	Network Fundamentals + CompTIA Security+ SY0-701 — 12 sessions of 3 hours
Strand	Security architecture (sessions 8–9)
Prerequisites	Sessions 1–3 network foundation; session 5 threats
Assessment	HTML learning quiz (20 questions) + Wayground quiz (15 questions)
Exam link	Domain 3 architecture objectives: segmentation, appliances, remote access, secure protocols

Session 8 coverage

ID	Area	What is covered
8A	Architecture	Tiered, spine-leaf, SMB; VLANs, DMZ, micro-segmentation; trust zones and zero trust
8B	Security Appliances	Firewalls, IDS/IPS, proxy, WAF, load balancer; hardening, NAC and 802.1X
8C	Secure Remote Access	VPN flavours, RDP controls, ZTNA, SASE, jump servers and bastion hosts
8D	Secure Protocols	TLS, HTTPS, IPsec, SSH, S/MIME, DNSSEC, HSTS, certificate validation
8E	Physical, Deception & Audit	Rooms and locks, surveillance, honeypots and honeynets, audits and regimes
8F	Segmentation & Special Environments	Segmentation principles, IoT, operational technology, the Purdue model
8G	Guests & Personal Devices	Guest VLANs, captive portals, BYOD, mobile device management
8H	Key Management & Content Control	The key lifecycle, key stores, ACLs, URL and DNS filtering, DLP, inspection

How this session maps to the course

Session 2's VLANs and DMZ become containment controls here. Session 6's cryptography returns as key management, and session 3's SSH returns in the protocol toolkit. Session 11 (security operations) will monitor the appliances and filters you place today. Every quiz question is tagged with its topic area (e.g. 8B Security Appliances).